

CS 260: Foundations of Data Science

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Fall 2026



HAVERFORD
COLLEGE

Admin

- **Lab 2** grades & feedback posted on Moodle
- **Study guide & practice midterm** posted
- I will be away for a conference Sep 26 - Oct 2
 - **Office hours**: 11:30-1:30pm Friday (Sep 25)
 - **Guest lecture** + midterm review on Sep 29
 - **No lab** on Sep 30
 - **Midterm 1**: in class 10/01

Midterm 1 Notes

- You may use one letter page (front and back) “study sheet”, handwritten, created by you
- You may also use a regular calculator
- Outside of your “study sheet” and calculator, **no other notes or resources**
- **As per the Honor Code, all work must be your own**

Why do we have exams?

- Process of synthesizing the material on your own is essential
- Preparing the “study sheet” is designed to facilitate that process

Outline for today

- Evaluation Metrics
 - Confusion matrices
 - Precision and recall
 - ROC curves
- Review

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Goals of Evaluation

- Think about what metrics are important for the problem at hand
- Compare different methods or models on the same problem
- Common set of tools that other researchers/users can understand

Training and Testing

(high-level idea)

- **Separate** data into “**train**” and “**test**”
 - n = num training examples
 - m = num testing examples
- **Fit** (create) the model using **training data**
 - e.g. sea_ice_1979-2012.csv
- **Evaluate** the model using **testing data**
 - e.g. sea_ice_2013-2020.csv

Confusion Matrices

		Predicted class		
		Negative	Positive	
True class	Negative	True negative (TN)	False positive (FP) "false alarm"	N
	Positive	False negative (FN) "miss"	True positive (TP)	P
		N*	p*	

Precision:

$$\frac{TP}{(FP + TP)} = \frac{TP}{P^*}$$

Confusion Matrices

		Predicted class		
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True class	Negative	True negative (TN)	False positive (FP) "false alarm"	N
	Positive	False negative (FN) "miss"	True positive (TP)	P
		N*	p*	

Recall
(True Positive Rate):

$$\frac{TP}{(FN + TP)} = \frac{TP}{P}$$

Confusion Matrices

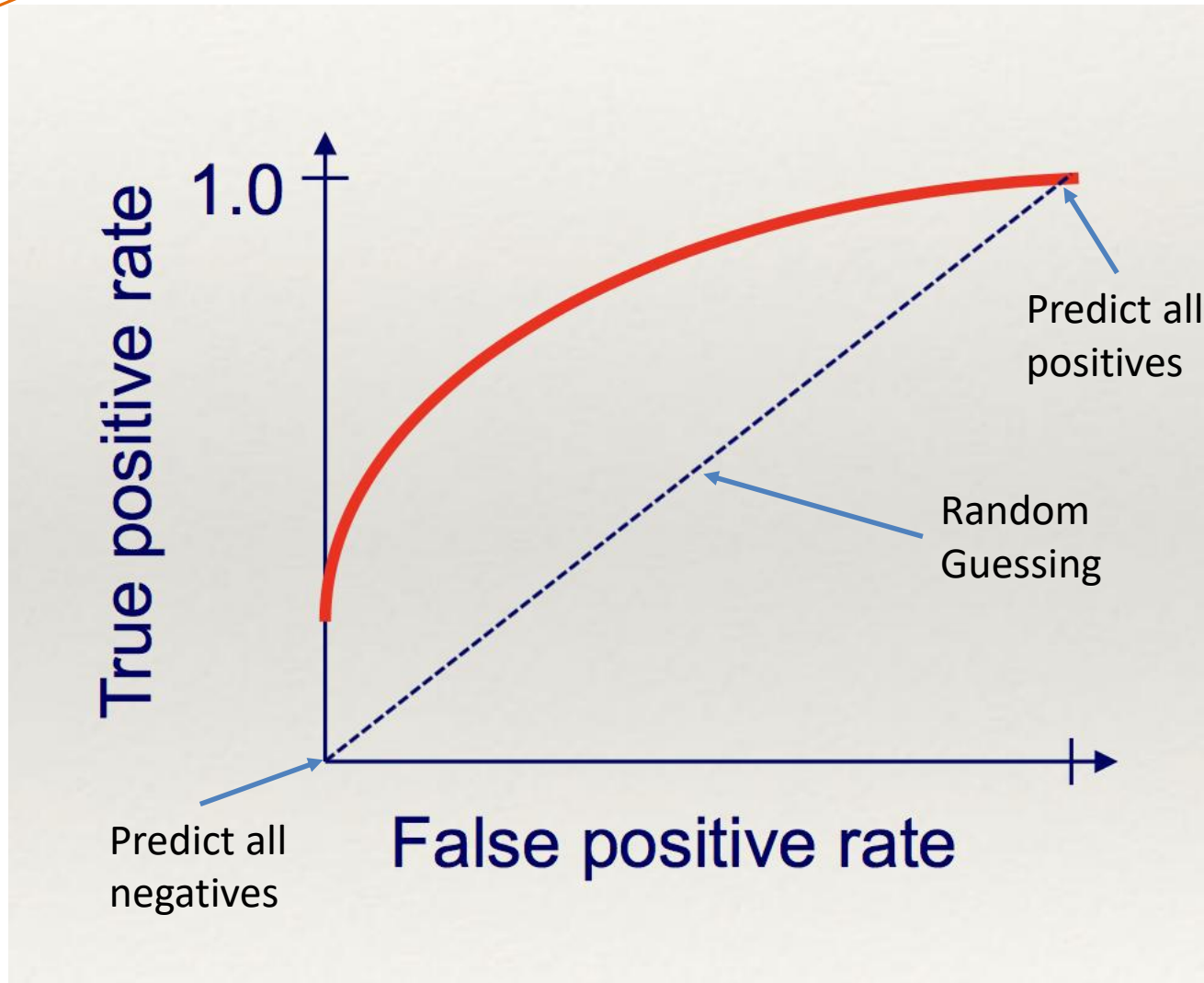
		Predicted class		
		Negative	Positive	
True class	Negative	True negative (TN)	False positive (FP) "false alarm"	N
	Positive	False negative (FN) "miss"	True positive (TP)	P
		N*	p*	

False Positive Rate:

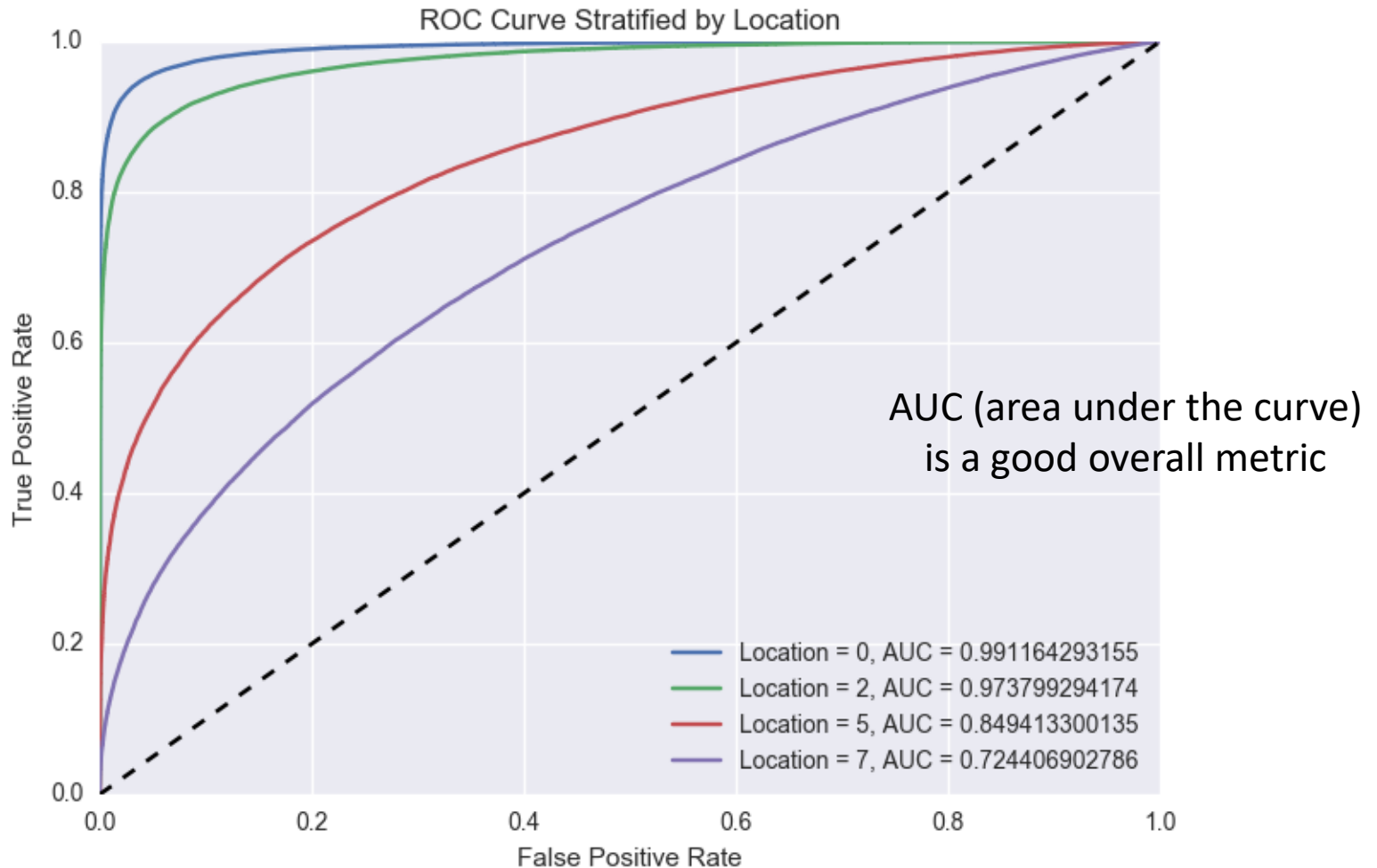
$$\frac{FP}{(TN + FP)} = \frac{FP}{N}$$

ROC curve (Receiver Operating Characteristic)

More history here!



ROC curve example: comparing methods



Example of a ROC curve

Chan, Perrone, Spence, Jenkins, Mathieson, Song

How to get an ROC curve for probabilistic methods?

- Usually we use 0.5 as a threshold for binary classification
- Vary the threshold! (i.e., choose 0, 0.1, 0.2,...)
- If $P(y = 1|x) \geq 0.2$, classify as 1 (positive)
- If $P(y = 1|x) < 0.2$, classify as 0 (negative)

Handout 8

Handout 8

	-	+
-	77	3
+	13	7

N=80

P=20

$N^*=90$ $P^*=10$

Precision=7/10

Recall=7/20=0.35

FPR=3/80

	-	+
-	68	12
+	2	18

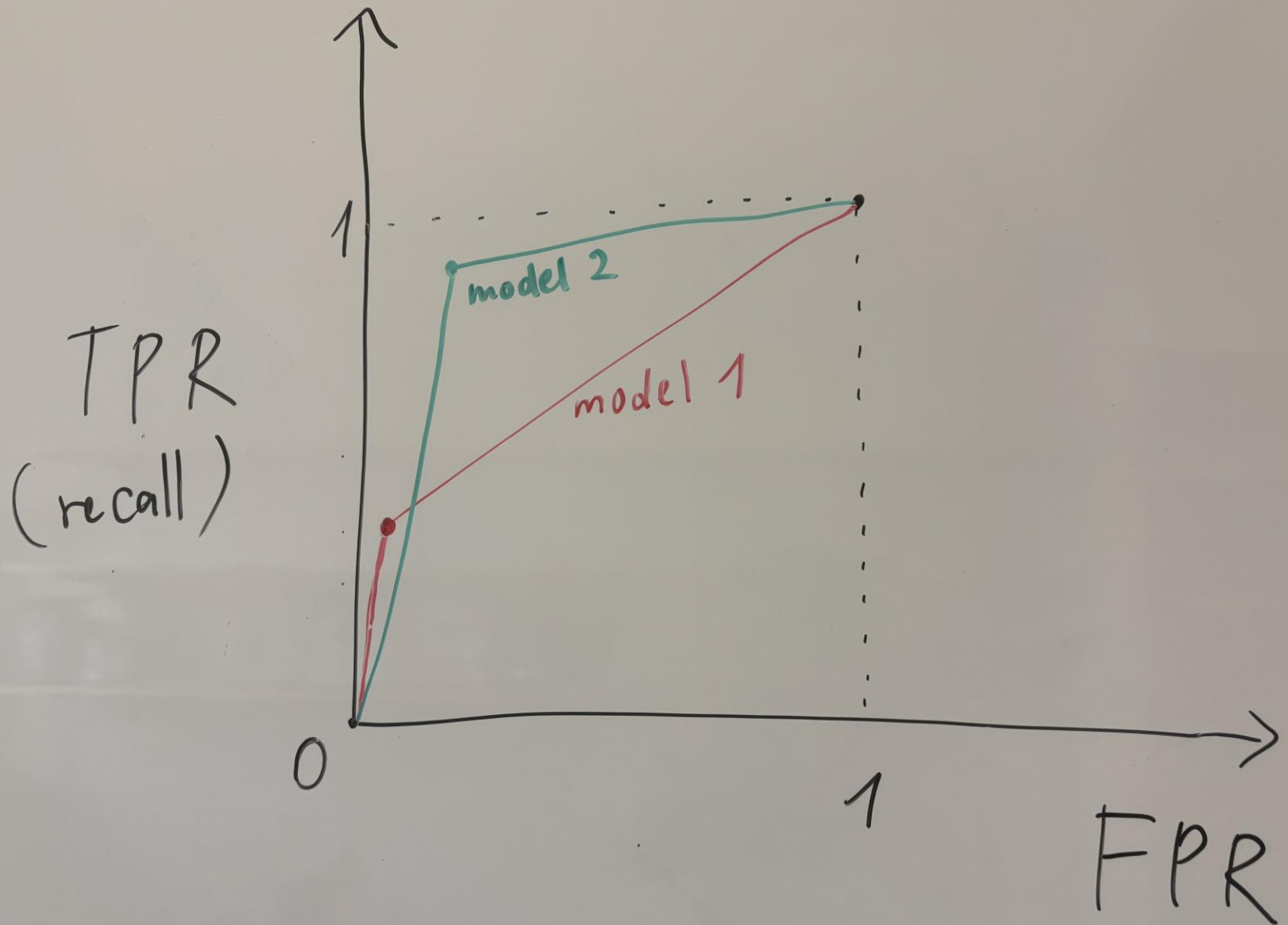
P=20

$P^*=30$

TPR = $18/20 = 0.9$

FPR = $12/80 = 0.15$

Handout 8



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Lab 2: not posted online

Multiple linear regression

Model: $h_{\vec{w}}(\vec{x}) = \sum_{j=0}^p w_j x_j = w_0 x_0 + w_1 x_1 + w_2 x_2 + \cdots + w_p x_p = \vec{w} \cdot \vec{x}$

Predict: $X\vec{w} = \hat{\vec{y}}$

Minimize cost:

“any” regression problem

$$\begin{aligned} J(\vec{w}) &= \frac{1}{2} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \\ &= \frac{1}{2} (\vec{y} - \hat{\vec{y}}) \cdot (\vec{y} - \hat{\vec{y}}) \\ &= \frac{1}{2} (\hat{\vec{y}} - X\vec{w}) \cdot (\hat{\vec{y}} - X\vec{w}) \end{aligned}$$

A handwritten equation on a chalkboard illustrating the matrix form of the linear regression model. It shows a matrix X of size $n \times (p+1)$ multiplied by a vector $\vec{w}$ of size $(p+1) \times 1$, resulting in a vector $\hat{\vec{y}}$ of size $n \times 1$. The matrix X is shown with its first row $[x_{i0} \ x_{i1} \ \dots \ x_{ip}]$ circled. The vector $\vec{w}$ is shown with its elements $w_0, w_1, \dots, w_p$ listed vertically. The resulting vector $\hat{\vec{y}}$ has its first element $\hat{y}_i$ circled. The equation is written as $X \vec{w} = \hat{\vec{y}}$.

Multiple linear regression vs. polynomial regression

Multiple linear regression ($p > 1$):

$$h_{\vec{w}}(\vec{x}) = w_0 + w_1x_1 + w_2x_2 + \cdots + w_px_p$$

Polynomial regression ($p=1$):

Note: d means degree

$$h_{\vec{w}}(\vec{x}) = w_0 + w_1x + w_2x^2 + \cdots + w_dx^d$$


"simple" linear
regression

$$X_d = \begin{bmatrix} 1 & x_1 & x_1^2 & x_1^3 & \cdots & x_1^d \\ 1 & & & & & \\ 1 & & & & & \\ \vdots & & & & & \\ 1 & & & & & \end{bmatrix} \quad \leftarrow \vec{x}_1$$

SGD with time varying α

α is the “learning rate” or “step size”

SGD pseudocode:

$t = 1$

While not converged:

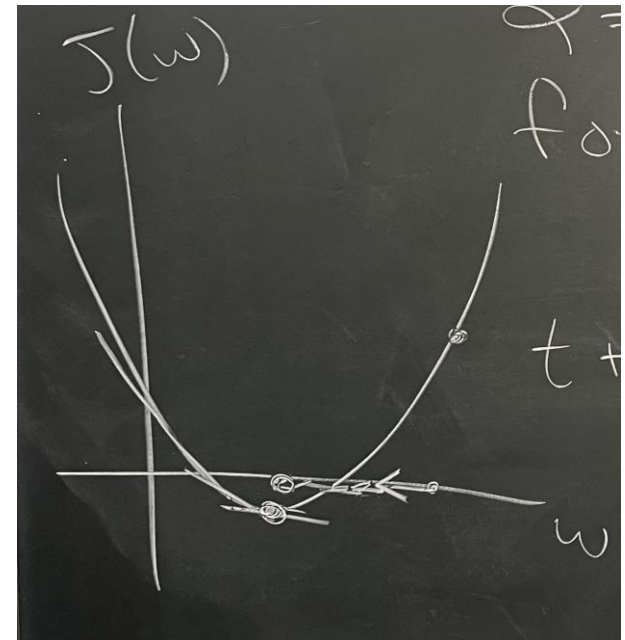
$$\alpha = \frac{1}{t}$$

for $i = 1 \cdots n$

$$\vec{w} \leftarrow \vec{w} - \alpha \underbrace{(h_{\vec{w}}(\vec{x}_i) - y_i)}_{\text{derivative of } J \text{ with respect to } \vec{x}_i} \vec{x}_i$$

$t += 1$

derivative of J
with respect to $\vec{x}_i$



Matrix/Vector form of SGD

while not converged:

shuffle the data (stochastic)

for $i = 1, 2, 3 \dots n$:

$$\begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_p \end{bmatrix} \leftarrow \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_p \end{bmatrix} - \alpha (\underbrace{h_{\vec{w}}(\vec{x}_i)}_{\text{pred}} - \underbrace{y_i}_{\text{truth}}) \begin{bmatrix} x_{i0} \\ x_{i1} \\ \vdots \\ x_{ip} \end{bmatrix}$$

Convergence:

$$| \text{new_cost} - \text{old_cost} | < \varepsilon$$